

Session 3 — Unit 1 Review: Vectors, Motion Graphs & Technical Drawings

Duration: 45 minutes

Continues from: Session 2, 2D Kinematics (projectile motion). Session 1 covered 1D kinematics and the SUVAT equations.

Why now: the school's **Unit 1 Review** (32 multiple-choice questions) covers vectors and scalars, technical drawings, 1D kinematics, acceleration and motion graphs, and 2D kinematics. Sessions 1–2 already covered the kinematics. Today fills the three gaps (**vectors done formally, motion graphs, technical drawings**) and finishes with an exam-style drill.

Grade-9 foundation: Ch4 *Describing Motion Around Us* (distance–time and speed–time graphs, §4.2).

Grade-11 depth: NCERT Class 11 Ch3 *Motion in a Plane* (§3.2 scalars and vectors, head-to-tail addition, §3.5 resolution into components), Ch2 *Motion in a Straight Line* (slope and area of x–t and v–t graphs; kinematic equations derived from v–t area). Also Glencoe *Physics: Principles and Problems* §2.3 (position–time graphs) and Hewitt *Conceptual Physics* Ch6 (vectors).

Not in any textbook in the vault: technical drawing types and scaling are engineering-drawing content from the school's PhysEng course. That section uses standard drafting conventions and is marked as such.

Units note: the school's answer choices use $g = 9.81 \text{ m/s}^2$ (Session 2 used 9.8). Use 9.81 from now on.

Lesson Plan

Objectives. By the end of this session the student should be able to:

1. Classify a quantity as scalar or vector, and tell distance from displacement and speed from velocity.
2. Add perpendicular vectors, then state the resultant's direction as a **standard-position angle (0°–360°, measured counter-clockwise from east)**. This is the format the school uses, e.g. "22.36 m at 297°".
3. Read position–time and velocity–time graphs: slope, area, and what each shape means.
4. Identify orthographic, isometric, oblique and perspective drawings, and apply a scale factor.

Time	Segment
0–4 min	Warm-up: two projectile recall questions
4–14 min	Scalars vs vectors, resultant + direction angle, components
14–26 min	Acceleration and motion graphs (slope, area, shapes)
26–34 min	Technical drawings and scaling
34–43 min	Exam-style rapid drill (mixed, timed)
43–45 min	Wrap-up: assign the practice test and the homework problems below

Pacing note. If you run behind, shorten the component-addition example in the vectors section (the test only needs perpendicular vectors) before touching anything else. Do not cut the direction-angle rule or the graphs section. Those are where the test's trickiest distractors live.

Warm-up (4 min)

1. Ball A is dropped from rest. At the same instant, Ball B is launched horizontally at 8.0 m/s from the same height. Which lands first? **Same time.** Vertical motion is independent of horizontal motion.

Both start with zero vertical velocity and both fall with g , so they take identical times. This is the Session 2 core idea and it's on his worksheet.

2. A marble rolls off a 1.25 m table. How long is it in the air? $t = \sqrt{2h/g} = \sqrt{2 \times 1.25 / 9.81} = \sqrt{0.2548} = 0.50 \text{ s}$. The 3.2 m/s horizontal speed doesn't matter for the time.

Part 1 — Scalars vs Vectors (10 min)

Definitions (NCERT Class 11 §3.2)

- **Scalar quantity:** specified completely by a single number (its magnitude) with the proper unit. Examples: distance, speed, time, mass, volume, temperature, energy.
- **Vector quantity:** has both a **magnitude and a direction**, and obeys the triangle (head-to-tail) law of addition. Examples: displacement, velocity, acceleration, force.

The two pairs that get tested most:

Scalar	Vector partner	Difference
Distance: total path length	Displacement: straight line from start to finish, with direction	Walk 30 m E then 40 m N: distance = 70 m, displacement = 50 m
Speed = distance $\div$ time	Velocity = displacement $\div$ time	Same walk in 50 s: average speed 1.4 m/s, average velocity 1.0 m/s

NCERT puts it this way: average speed is always **greater than or equal to** the magnitude of average velocity. They are equal only for straight-line motion that never turns back.

Adding perpendicular vectors: magnitude

Place the vectors **head to tail**. The resultant R runs from the tail of the first to the head of the last. If the two vectors are perpendicular they form a right triangle with R as the hypotenuse, so by Pythagoras:

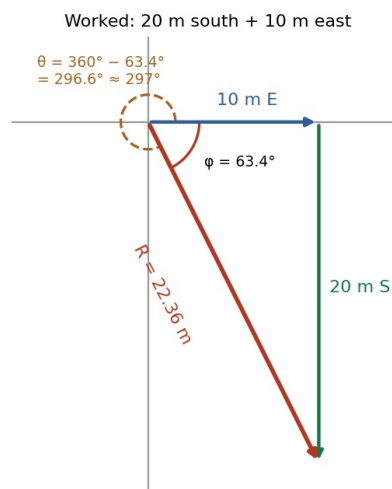
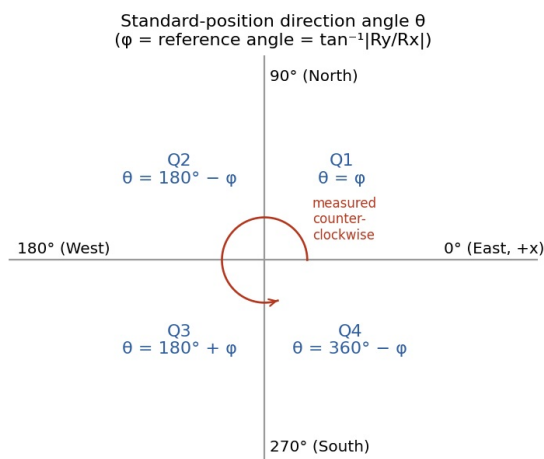
$$R = \sqrt{R_x^2 + R_y^2}$$

The direction angle: this is where marks are lost

The school states direction as a **standard-position angle θ** . It is measured **counter-clockwise from the +x axis (east)**, from 0° to 360° . The recipe:

1. Assign signs: east and north are **+**, west and south are **−**. So R_x is **+** for east, **−** for west; R_y is **+** for north, **−** for south.
2. Find the **reference angle** $\phi = \tan^{-1}(|R_y| / |R_x|)$. This is always between 0° and 90° .
3. Use the quadrant to convert ϕ into θ :

Quadrant	Signs (R_x, R_y)	θ
Q1 (north-east)	(+, +)	$\theta = \phi$
Q2 (north-west)	(−, +)	$\theta = 180^\circ - \phi$
Q3 (south-west)	(−, −)	$\theta = 180^\circ + \phi$
Q4 (south-east)	(+, −)	$\theta = 360^\circ - \phi$



Standard-position angle and the worked example

Why the quadrant step matters. For “20 m south and 10 m east”, $\phi = \tan^{-1}(20/10) = 63.4^\circ$, and **63° is one of the wrong answer choices**. The resultant points south-east (Q4), so $\theta = 360^\circ - 63.4^\circ = 296.6^\circ \approx 297^\circ$. The calculator only gives ϕ . Sketching the arrows gives the quadrant. Always sketch first.

Worked Example 1. A drone flies 12 m north, then 5 m west. Find the resultant displacement.

- Components: $R_x = -5 \text{ m}$ (west), $R_y = +12 \text{ m}$ (north), which is quadrant Q2.
- Magnitude: $R = \sqrt{5^2 + 12^2} = \sqrt{169} = 13 \text{ m}$
- Reference angle: $\phi = \tan^{-1}(12/5) = 67.4^\circ$
- Standard angle (Q2): $\theta = 180^\circ - 67.4^\circ = 112.6^\circ$
- Compass form, if a question asks for it: 22.6° west of north, since $\tan^{-1}(5/12) = 22.6^\circ$.

Answer: 13 m at 112.6°.

Resolving a vector into components (NCERT §3.5, Hewitt 6.6)

The adding process runs in reverse. A vector A at standard angle θ is the hypotenuse of a right triangle whose legs are its x- and y-components:

$$A_x = A \cos \theta \quad A_y = A \sin \theta$$

These come straight from SOH-CAH-TOA. They are exactly how Session 2 split a launch velocity into $v_0 \cos \theta$ and $v_0 \sin \theta$. When θ is a standard-position angle, the **signs come out automatically**. For example, 15 m/s at 210° gives $v_x = 15 \cos 210^\circ = -12.99 \text{ m/s}$ and $v_y = 15 \sin 210^\circ = -7.50 \text{ m/s}$, which points west and south, as it should.

To add vectors that are **not** perpendicular: resolve each one, add all the x-components, add all the y-components, then use Pythagoras and the quadrant rule on the totals. Practice Problem 4 below does this.

Part 2 — Acceleration & Motion Graphs (12 min)

Acceleration

$a = \Delta v / \Delta t = (v - u) / t$, in m/s^2 . This is the Session 1 equation $v = u + at$, rearranged.

- If a has the **same sign** as v, the object is **speeding up**. If it has the **opposite sign**, it is **slowing down**. “Negative acceleration” does not always mean slowing down.
- **Free fall:** $a = g = 9.81 \text{ m/s}^2$, downward, whatever the mass (no air resistance).

Rule 1: slope of a position-time graph = velocity

Derivation: slope = rise $\div$ run = $\Delta x \div \Delta t$, and $\Delta x \div \Delta t$ is the **definition** of average velocity. On a curved x-t graph, the velocity at an instant is the slope of the tangent at that point (NCERT Ch2 summary, point 2).

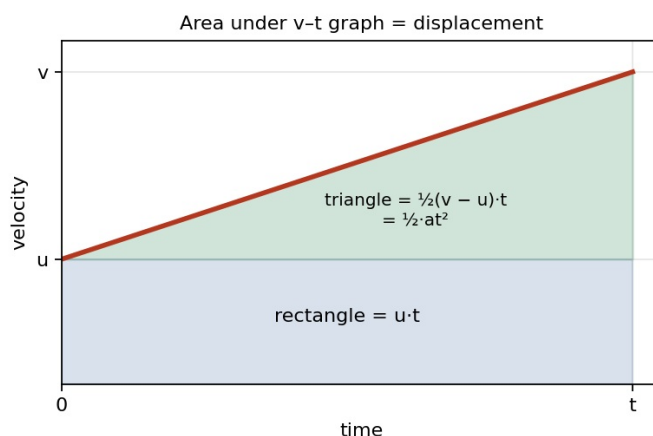
Rule 2: slope of a velocity-time graph = acceleration

Same logic: slope = $\Delta v \div \Delta t$, which is the definition of acceleration (NCERT Ch2 summary, point 4).

Rule 3: area under a velocity-time graph = displacement

For constant velocity the v-t graph is a flat line, so the area is a rectangle of height v and width t. **Area = $v \times t = x$.**

For uniform acceleration from u to v (NCERT §2.4, Fig 2.5), split the area into two pieces:

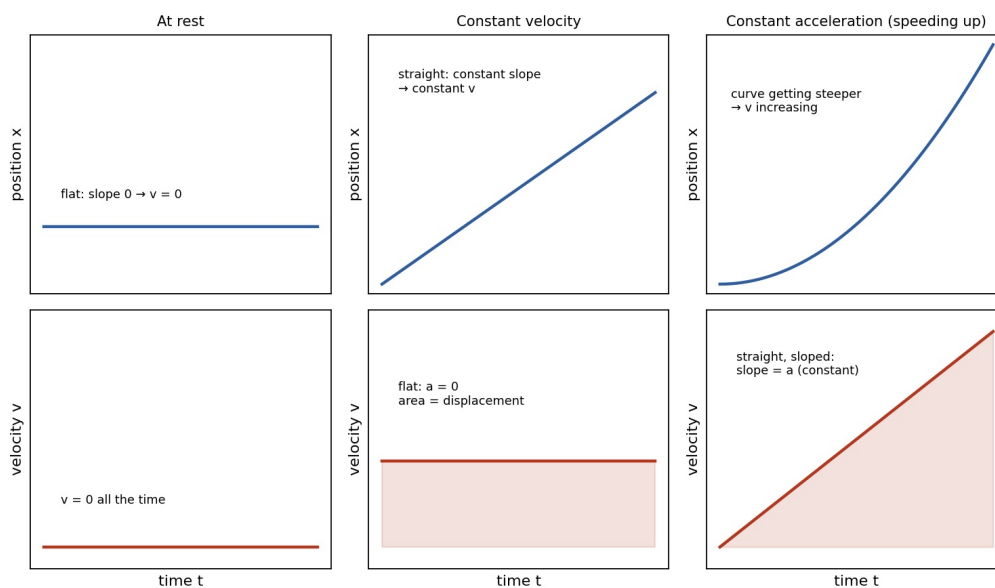


Area under the v-t graph

- Rectangle: $u \times t$
- Triangle: $\frac{1}{2} \times (v - u) \times t$. Since $v - u = at$ (from the definition of a), the triangle is $\frac{1}{2} \times at \times t = \frac{1}{2}at^2$.
- Total: $x = ut + \frac{1}{2}at^2$

That is exactly the Session 1 SUVAT equation, now **derived from a graph**. Similarly, the **area under an acceleration-time graph = change in velocity**, because $a \times t = \Delta v$.

What the shapes mean (NCERT Ch2 summary, point 4)



x-t and v-t graphs for the three basic motions

Motion	x-t graph	v-t graph
At rest	Horizontal line	On the time axis ($v = 0$)
Constant velocity	Straight sloped line	Horizontal line ($a = 0$)
Constant acceleration	Parabola (curve)	Straight sloped line
Moving backward	Line sloping downward	Below the time axis

Trap to flag. The graph on the school review has **distance** on the y-axis and a **straight** line. Straight means constant slope, so constant velocity. It is a position graph, not a velocity graph. Reading the slope identifies the scenario: about 75 m in 9 s, which is 8.33 m/s.

Worked Example 2 (v-t graph). A car speeds up from rest to 12 m/s in 4 s, cruises at 12 m/s for 6 s, then brakes to rest in 3 s.

- Accelerations (slopes): $a_1 = (12 - 0)/4 = 3 \text{ m/s}^2$, $a_2 = 0$, $a_3 = (0 - 12)/3 = -4 \text{ m/s}^2$
- Displacement (area): triangle $\frac{1}{2} \times 4 \times 12 = 24 \text{ m}$, plus rectangle $6 \times 12 = 72 \text{ m}$, plus triangle $\frac{1}{2} \times 3 \times 12 = 18 \text{ m}$, for a **total of 114 m**
- Average velocity: $114 \text{ m} \div 13 \text{ s} = 8.77 \text{ m/s}$

Part 3 — Technical Drawings & Scaling (8 min)

Standard engineering-drawing conventions from the school's PhysEng course. None of the vault textbooks cover this, so it is supplementary, not textbook-verified.

Why scale?

Real objects are often too big (a bridge, a building) or too small (a watch gear) to draw at true size. Scaling shrinks or enlarges **every** dimension by the same factor, so all proportions stay true and real measurements can still be read off the drawing. In the review's words, the purpose is "to fit a large image/object onto a smaller piece of paper."

Scale notation is drawing : actual .

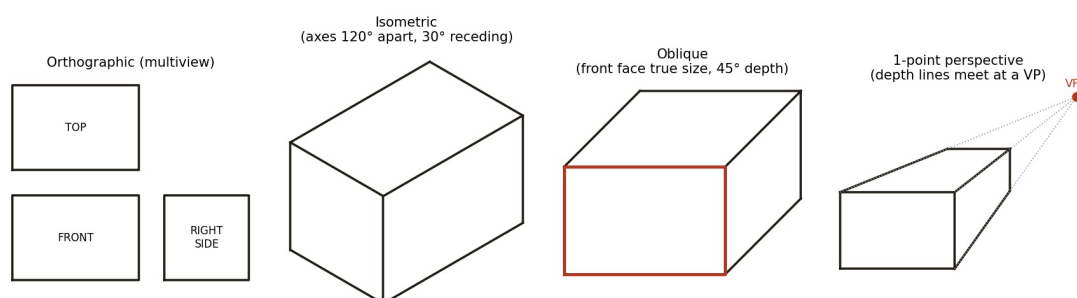
- **1:1**: full size.
- **1:2**: half size (a reduction). Divide the real lengths by 2.
- **2:1**: double size (an enlargement). Multiply the real lengths by 2.
- General rule: **drawing length = actual length $\times$ (first number $\div$ second number).**

Worked Example 3. A triangle with sides 6, 9 and 14 cm:

- at **3:1** it is drawn with sides 18, 27 and 42 cm ($\times 3$)
- at **1:3** it is drawn with sides 2, 3 and 4.67 cm ($\div 3$)

In reverse: a 4.5 cm line on a 1:50 plan is $4.5 \times 50 = 225 \text{ cm} = 2.25 \text{ m}$ in real life.

The four drawing types



The same block drawn four ways

Type	What it looks like	Key identifying fact
Orthographic (multiview)	Separate flat 2D views: front, top, right side	Each view shows its face in true size and shape . Standard layout: top view above the front view, right-side view to its right. Hidden edges are dashed.
Isometric	Pictorial 3D, corner facing you	The three axes are 120° apart , with receding lines at 30° to horizontal. Lengths along the axes are true, but no face appears in its true shape.
Oblique	Pictorial 3D, front face flat to you	The front face is drawn in true shape and proportion (the “frontal lines with true proportions and relations”). Depth lines recede at an angle, usually 45°.
Perspective	Most realistic, like a photo	Parallel depth lines converge to vanishing point(s) (1-point, 2-point). Looks real, but you can’t measure true sizes from it.

Distractor to flag: AutoCAD is **software** for making drawings (CAD = computer-aided design). It is not a type of drawing.

Two refinements of oblique, in case they come up: **cavalier** oblique draws depth at full length, and **cabinet** oblique draws depth at half length, which looks less stretched.

Part 4 — Exam-Style Rapid Drill (9 min)

Give about 60–90 seconds each, answered aloud or on the whiteboard. Answers are for the tutor.

1. A runner covers 100 m in 12.5 s at constant velocity. Velocity? → $100/12.5 = 8.0 \text{ m/s}$
2. Which is a scalar: displacement, velocity, speed, or acceleration? → **Speed**
3. A hiker goes 8 m west and 6 m south. Resultant? → $\sqrt{64 + 36} = 10 \text{ m}$; $\phi = \tan^{-1}(6/8) = 36.9^\circ$; Q3, so $\theta = 180^\circ + 36.9^\circ = 216.9^\circ$
4. The slope of a v–t graph is...? → **Acceleration**. The area under it? → **Displacement**
5. A drawing shows a 3 m wall as 6 cm. What’s the scale? → $6 \text{ cm} : 300 \text{ cm} = 1:50$
6. A ball rolls off a 20 m roof and must land 6 m out. What horizontal speed does it need? → $t = \sqrt{(2 \times 20 / 9.81)} = 2.02 \text{ s}$, then $v = 6 / 2.02 = 2.97 \text{ m/s}$. The horizontal distance fixes the speed, so “it doesn’t matter” is wrong.
7. Which drawing type shows the front face in true proportion, with depth receding at 45°? → **Oblique**

Wrap-up (2 min)

- One-line summary for him: *vectors need a direction angle, so sketch first and apply the quadrant rule; slope of x–t is v, slope of v–t is a, area under v–t is displacement; oblique = true front face, isometric = 120° axes, perspective = vanishing points.*
- Homework: the **Unit 1 Practice Test** (separate file, 32 questions in the same format as the school’s review) plus Practice Problems 1–11 below.

Practice Problems

Use $g = 9.81 \text{ m/s}^2$. Angles are standard-position (counter-clockwise from east) unless stated.

1. **Scalar or vector?** Classify each: 5 kg · 20 m/s north · 30 s · 9.81 m/s² downward · 12 N left · 40 km · 25 °C.
 2. **Distance vs displacement:** A hiker walks 30 m east then 40 m north in 50 s. Find (a) distance, (b) displacement (magnitude and angle), (c) average speed, (d) average velocity.
 3. **Resultant in Q3:** A boat moves 9 m west and 12 m south. Find the magnitude and standard-position direction of its displacement.
 4. **Adding non-perpendicular vectors:** A = 10 m at 30°, B = 6 m at 150°. Find A + B.
 5. **Resolving:** A velocity of 20 m/s at 330°. Find v_x and v_y , and say which compass quadrant it points into.
 6. **Reading a position-time graph:** A runner goes from 0 to 40 m in 8 s, stays at 40 m for 4 s, then returns to the 10 m mark by $t = 18 \text{ s}$ (all at constant speeds). Find the velocity in each stage, the total distance, the displacement, the average speed and the average velocity.
 7. **Braking (v-t area):** A car at 25 m/s brakes uniformly to rest in 5.0 s. Find its acceleration and stopping distance using the graph area, then check with a SUVAT equation.
 8. **Free fall:** A stone is dropped from a 20 m roof. Find the fall time and impact speed.
 9. **Horizontal launch (fire-cushion type):** From the same 20 m roof, a person leaps horizontally toward a cushion 6.0 m from the wall. What horizontal speed is needed?
 10. **Scaling:** Draw a 6, 9, 14 cm triangle at (a) 3:1 and (b) 1:3. What length is each side on paper?
 11. **Drawing ID:** Name the drawing type: (a) three flat views, front/top/side; (b) 3D view whose depth lines meet at a point on the horizon; (c) 3D view with axes 120° apart; (d) 3D view with a true-shape front face and 45° depth lines.
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Answer Key — Full Step-by-Step Solutions

1. Scalar or vector

- Step 1: A vector must state a direction. A scalar is a size only.
- Step 2: The quantities with directions are 20 m/s north (velocity), 9.81 m/s² downward (acceleration) and 12 N left (force).
- Step 3: The rest have no direction: 5 kg (mass), 30 s (time), 40 km (distance), 25 °C (temperature).

Answer: vectors = 20 m/s N, 9.81 m/s² down, 12 N left. Scalars = 5 kg, 30 s, 40 km, 25 °C.

2. Distance vs displacement

- Step 1 (distance, the total path): $30 + 40 = 70 \text{ m}$.
- Step 2 (displacement magnitude): the legs are perpendicular, so $\sqrt{(30^2 + 40^2)} = \sqrt{2500} = 50 \text{ m}$.
- Step 3 (direction): $R_x = +30$ and $R_y = +40$, which is Q1. $\theta = \tan^{-1}(40/30) = 53.1^\circ$.
- Step 4 (average speed): $70 / 50 = 1.4 \text{ m/s}$.
- Step 5 (average velocity): $50 / 50 = 1.0 \text{ m/s}$ at 53.1° .

Answer: 70 m; 50 m at 53.1°; 1.4 m/s; 1.0 m/s at 53.1°.

3. Resultant in Q3

- Step 1: $R_x = -9 \text{ m}$ (west), $R_y = -12 \text{ m}$ (south), which is quadrant Q3.
- Step 2: $R = \sqrt{(9^2 + 12^2)} = \sqrt{225} = 15 \text{ m}$.
- Step 3: $\phi = \tan^{-1}(12/9) = 53.1^\circ$.
- Step 4 (Q3): $\theta = 180^\circ + 53.1^\circ = 233.1^\circ$.

Answer: 15 m at 233.1°.

4. Adding non-perpendicular vectors

- Step 1 (resolve A): $A_x = 10\cos 30^\circ = 8.66 \text{ m}$, $A_y = 10\sin 30^\circ = 5.00 \text{ m}$.
- Step 2 (resolve B): $B_x = 6\cos 150^\circ = -5.20 \text{ m}$, $B_y = 6\sin 150^\circ = 3.00 \text{ m}$.
- Step 3 (add components): $R_x = 8.66 - 5.20 = 3.46 \text{ m}$, $R_y = 5.00 + 3.00 = 8.00 \text{ m}$.
- Step 4: $R = \sqrt{(3.46^2 + 8.00^2)} = \sqrt{(12.0 + 64.0)} = \sqrt{76.0} = 8.72 \text{ m}$.
- Step 5: both components are positive (Q1), so $\theta = \tan^{-1}(8.00/3.46) = 66.6^\circ$.

Answer: 8.72 m at 66.6°.

5. Resolving

- Step 1: $v_x = 20\cos 330^\circ = 20 \times 0.866 = 17.3 \text{ m/s}$.
- Step 2: $v_y = 20\sin 330^\circ = 20 \times (-0.5) = -10.0 \text{ m/s}$.
- Step 3: + x and - y means east and south, i.e. Q4.

Answer: $v_x = +17.3 \text{ m/s}$, $v_y = -10.0 \text{ m/s}$, pointing south-east.

6. Reading a position-time graph

- Stage 1 velocity (slope): $(40 - 0) / 8 = 5.0 \text{ m/s}$.
- Stage 2: the position doesn't change, so the slope is 0 and $v = 0$ (at rest).
- Stage 3: $(10 - 40) / (18 - 12) = -30 / 6 = -5.0 \text{ m/s}$. The minus sign means moving back toward the start.
- Total distance: $40 \text{ m out} + 30 \text{ m back} = 70 \text{ m}$.
- Displacement: $\text{final} - \text{initial} = 10 - 0 = 10 \text{ m}$.
- Average speed: $70 / 18 = 3.89 \text{ m/s}$. Average velocity: $10 / 18 = 0.56 \text{ m/s}$.

Answer: 5.0, 0, -5.0 m/s; 70 m; 10 m; 3.89 m/s; 0.56 m/s.

7. Braking (v-t area)

- Step 1 (acceleration = slope): $a = (0 - 25) / 5.0 = -5.0 \text{ m/s}^2$.
- Step 2 (distance = area of the triangle): $\frac{1}{2} \times 5.0 \times 25 = 62.5 \text{ m}$.
- Step 3 (check with $v^2 = u^2 + 2as$): $0 = 25^2 + 2(-5.0)s$, so $s = 625 / 10 = 62.5 \text{ m}$. ✓

Answer: $a = -5.0 \text{ m/s}^2$, stopping distance = 62.5 m.

8. Free fall

- Step 1: it starts from rest, so $h = \frac{1}{2}gt^2$ and $t = \sqrt{(2h/g)} = \sqrt{(40 / 9.81)} = \sqrt{4.08} = 2.02 \text{ s}$.
- Step 2: $v = gt = 9.81 \times 2.02 = 19.8 \text{ m/s}$.
- Step 3 (check): $v = \sqrt{(2gh)} = \sqrt{(2 \times 9.81 \times 20)} = \sqrt{392.4} = 19.8 \text{ m/s}$. ✓

Answer: 2.02 s, 19.8 m/s.

9. Horizontal launch

- Step 1: the fall time depends only on height. From Q8, $t = 2.02 \text{ s}$.
- Step 2: horizontal velocity is constant, so $v = x / t = 6.0 / 2.02 = 2.97 \text{ m/s}$.

Answer: about 2.97 m/s. The speed matters: too slow and they land short of the cushion.

10. Scaling

- Step 1: 3:1 means drawing = actual $\times 3$, giving 18, 27, 42 cm.
- Step 2: 1:3 means drawing = actual $\div 3$, giving 2, 3, 4.67 cm.

Answer: (a) 18, 27, 42 cm; (b) 2, 3, 4.67 cm.

11. Drawing ID

Answer: (a) orthographic / multiview; (b) perspective; (c) isometric; (d) oblique.

Notes for next session

After the Unit 1 test, the plan returns to **Forces (Newton's laws)**. That was Session 3 in the master plan and is now Session 4. If the test results show a weak area, start Session 4 with a 5-minute fix on it. The most likely candidates are the quadrant rule for direction angles and reading x-t versus v-t graphs.